

REFERENCE SOLUTION ARCHITECTURE

Next-Generation Document Processing Engine

*Powered by Azure AI Vision, Azure AI Document Intelligence, Azure OpenAI,
Microsoft Fabric, and the Microsoft workflow stack*

Conduent Industry Solutions

Volume 2 of 3 — Solution Map and Conversion Plan

1. Solution Overview

This document defines the reference architecture for Conduent's next-generation document processing engine — a multimodal, AI-native pipeline that ingests cargo and shipping documents, classifies them, harvests structured and unstructured content, persists data into a governed lake, generates real-time insights, and routes work intelligently between automated processing, Conduent operators, and customer subject matter experts.

The architecture is built entirely on Microsoft Azure AI services and the broader Microsoft cloud stack, allowing Conduent to deliver enterprise-grade reliability, security, compliance, and scale while preserving optionality on the foundation models that power document understanding.

Reference architecture at a glance

Stage	Capability
1. Capture	Multi-channel ingestion: scanner, mobile, email, SFTP, API, fax bridge
2. Pre-process	Azure AI Vision — deskew, denoise, orient, enhance, detect stamps and barcodes
3. Identify	Azure AI Document Intelligence classifier — selects extraction model per document type
4. Harvest	Document Intelligence custom + prebuilt models, augmented by Azure OpenAI multimodal reasoning
5. Validate	Cross-field, cross-document, and business-rule validation; PII detection via Purview / Content Safety
6. Persist	Microsoft Fabric / OneLake — bronze, silver, gold layers with full lineage
7. Insight	Power BI dashboards, Azure AI Search semantic index, anomaly detection
8. Route	Logic Apps and Power Automate — confidence-based routing to STP, assisted review, or SME adjudication
9. Engage	Microsoft Copilot Studio agents in Teams for Conduent operators and customer SMEs
10. Learn	SME corrections feed weekly retraining of custom Document Intelligence models

2. Component Roles and Capabilities

This section walks through each Microsoft technology in the solution and articulates the role it plays, the capabilities it brings, and the document types where it is most relevant.

2.1 Azure AI Vision

Azure AI Vision provides the image-level intelligence that prepares every captured document for higher-order understanding. It is the entry point of the pipeline.

Role

Image normalization, baseline OCR, object detection, and quality scoring.

Capabilities

- Image enhancement: deskew, denoise, orientation correction, contrast normalization
- Read API: production-grade OCR for printed and handwritten text in 100+ languages
- Object and seal detection: identifies stamps, signatures, logos, barcode regions, photographs of cargo
- Image quality scoring: gates poor captures back to the originator with specific guidance (re-photograph, flatten, increase light)
- Spatial analysis on bundled documents — separates a multi-page mobile capture into discrete documents

Document relevance

- Highest value on mobile-captured PoDs, weigh tickets, dock receipts, condition photos
- Critical pre-processing layer for legacy paper bills of lading and customs forms received via fax bridge

2.2 Azure AI Document Intelligence

Azure AI Document Intelligence (formerly Form Recognizer) is the workhorse of structured and semi-structured extraction. It combines prebuilt models for common document categories with custom neural models trained on Conduent's cargo-specific corpus.

Role

Document classification, layout analysis, key-value extraction, table extraction, and confidence scoring per field.

Capabilities

- Prebuilt models: invoices, receipts, IDs, business cards, contracts — applied where relevant
- Layout API: recognizes tables, selection marks, and reading order across complex multi-column documents

- Custom neural models: trained on weigh bills, AWBs, BOLs, customs declarations, dangerous-goods declarations, and packing lists
- Document classifier: routes incoming documents to the correct extractor — replaces brittle template-matching logic
- Per-field confidence scores feed downstream routing decisions
- Active-learning support: weekly retraining on SME-corrected examples

Document relevance

- Air waybills: AWB number, shipper, consignee, pieces, weight, dimensions, charges — custom model, 95%+ field accuracy target
- Bills of lading: container, vessel, voyage, ports, cargo description, marks and numbers — custom model
- Commercial invoices: line items, HS codes, country of origin, Incoterms, currency, totals — prebuilt invoice model + custom overlay
- Weigh bills: gross / tare / net weight, vehicle, route, scale ID, ticket number — custom model
- Customs declarations: tariff codes, declared values, duties — custom model with cross-validation against invoices

2.3 Azure OpenAI Service (Multimodal Foundation Models)

Azure OpenAI provides the reasoning, comprehension, and generation layer. It is invoked for tasks that exceed the structured-extraction remit of Document Intelligence — handling novel layouts, ambiguous content, cross-document validation, and natural-language outputs.

Role

Document understanding for unstructured and novel content; reasoning across fields, documents, and historical context; generation of summaries and next-best-action narratives.

Capabilities

- Multimodal vision-text reasoning over the full document image
- Few-shot extraction on novel layouts without retraining
- Cross-field validation: 'Does the HS code match the goods description?' 'Is the declared weight plausible given the dimensions?'
- Cross-document validation: invoice ↔ AWB ↔ BOL ↔ packing list reconciliation
- Multi-language understanding and normalization in a single pass
- Handwriting interpretation for inspector annotations and condition notes
- Natural-language summarization, exception narratives, and next-best-action recommendations
- Structured output via function-calling: returns clean JSON aligned to the canonical schema

Document relevance

- Highest value on PoDs with handwritten condition notes
- Dangerous-goods declarations requiring UN-number and packing-group reasoning
- Customs documents requiring cross-reference to invoices and BOLs
- Any novel layout the customer onboards before a custom model is trained

2.4 Azure AI Search

Azure AI Search builds the queryable knowledge layer that turns the document data lake into a live operational asset.

Role

Hybrid (vector + keyword) semantic search across all extracted content, providing the retrieval layer for operator copilots, customer self-service, and analytics.

Capabilities

- Vector embeddings of document content and extracted narratives
- Hybrid search combining semantic similarity with structured filters (date, customer, lane, doc type)
- Indexer integration with Fabric / OneLake — new documents become searchable within seconds of capture

- Powers retrieval-augmented generation (RAG) for the Copilot Studio agents

2.5 Microsoft Fabric and OneLake

Microsoft Fabric provides the unified analytics platform; OneLake is its single, organization-wide data lake. Together they replace the patchwork of staging databases, data warehouses, and reporting marts that legacy operations have accumulated.

Role

Single source of truth for all document, extraction, validation, and operational data — bronze, silver, and gold layers — with built-in lineage, governance, and analytics.

Capabilities

- Bronze layer: raw captured documents, capture metadata, ingestion provenance
- Silver layer: extracted fields with confidence scores, classifications, links between documents
- Gold layer: business-ready entities — shipment, consignment, billing event, exception
- Real-Time Intelligence for streaming ingestion — documents flow into the lake as they are captured
- Direct Lake mode for Power BI — dashboards query the lake without copy
- Native integration with Microsoft Purview for cataloging, lineage, classification, and access policy

2.6 Logic Apps and Power Automate

Logic Apps and Power Automate provide the workflow routing fabric that connects the AI components, the Conduent operations center, the customer SMEs, and the customer's downstream ERP/TMS systems.

Role

Workflow orchestration, business-rule routing, system-of-record integration, SLA management, and alerting.

Capabilities

- Confidence-based routing: high-confidence to STP, medium to assisted review, low to SME adjudication
- Business-rule routing: customs anomalies, sanctions hits, dangerous-goods flags routed to specialists
- Connectors into client ERP (SAP, Oracle, Microsoft Dynamics 365), TMS, customs systems, billing platforms
- SLA timers with proactive escalation
- Approval flows in Microsoft Teams for SME adjudication
- Alerts via Teams, email, and mobile push
- Auto-generated next-best-action narratives delivered alongside each routed work item

2.7 Microsoft Copilot Studio and Microsoft Teams

Copilot Studio brings purpose-built AI agents into the daily working environment of Conduent operators and customer SMEs — Microsoft Teams.

Role

Agent-driven user interfaces for triage, adjudication, customer self-service, and operational analytics.

Capabilities

- Operator copilot — assists Conduent triage and exception teams with conversational document Q&A and recommended actions
- SME copilot — provides customer-side adjudicators with full extraction context, validation history, and prior shipment lookups
- Customer self-service agent — embedded in the customer portal for shipment status, document inquiries, and invoice questions
- Reasoning trace exposed for auditability — every recommended action shows its source documents and validation steps

2.8 Microsoft Purview and Azure AI Content Safety

Governance is not an afterthought; it is woven through the architecture. Microsoft Purview catalogs the data lake; Azure AI Content Safety screens content at point of capture and generation.

Role

Compliance, governance, lineage, PII detection, and responsible-AI guardrails.

Capabilities

- Automatic PII classification on cargo manifests, customs forms, and invoices
- Sensitivity labels propagating from capture through analytics
- Lineage from raw document to gold-layer business entity
- Content Safety filtering on agent responses
- Audit trails for every extraction, validation, and SME adjudication

3. Document Type Coverage Matrix

The architecture is exercised across the full mix of cargo and shipping document types. The matrix below summarizes the primary capability mapping, indicating which Azure components contribute and the principal extracted entities.

Document type	Primary Azure components	Key extracted entities
Air waybills (AWB)	AI Vision + Document Intelligence (custom)	AWB number, shipper, consignee, pieces, weight, charges, flight, route
Bills of lading (BOL)	Document Intelligence (custom) + Azure OpenAI	Container, vessel, ports, cargo description, marks and numbers
Commercial invoices	Document Intelligence (prebuilt + custom) + Azure OpenAI	Line items, HS codes, country of origin, Incoterms, currency, totals
Weigh bills	AI Vision + Document Intelligence (custom)	Gross / tare / net weight, vehicle ID, route, scale ID, ticket number
Customs declarations	Document Intelligence (custom) + Azure OpenAI	Tariff codes, declared values, duties, regulatory references
Dangerous goods declarations	Document Intelligence (custom) + Azure OpenAI	UN number, packing group, hazard class, emergency contact
Proof of delivery (PoD)	AI Vision + Azure OpenAI	Delivery timestamp, signatures, condition notes, exceptions
Packing lists	Document Intelligence (custom)	SKU, quantity, weight, dimensions, package count
Certificates of origin	Document Intelligence (prebuilt) + Azure OpenAI	Issuer, exporter, consignee, country of origin, goods
Arrival notices and manifests	Document Intelligence (custom)	ETA, vessel, port, container list, consignee

4. Core Processing Capabilities

Beyond the per-document mechanics, the solution is organized around four cross-cutting capability domains that the customer's executive sponsors and operations leadership recognize as the value proposition of the modernized engine.

4.1 Document Identification

Every captured document, regardless of channel, is identified before extraction. The Document Intelligence classifier — supplemented by Azure OpenAI when the classifier confidence is low — assigns a document type and selects the appropriate extraction pipeline. This replaces the brittle template-matching layer that defines legacy OCR pipelines.

Outcome

- New layouts no longer break the pipeline — they are absorbed by the classifier and extracted using a fallback Azure OpenAI prompt
- Mis-classified documents are detected (low classifier confidence) and routed to SME for type confirmation, with the classifier learning from the correction
- Bundled multi-document captures are split automatically before processing

4.2 Data Harvesting

Each document type maps to an extraction pipeline that combines Document Intelligence custom or prebuilt models with Azure OpenAI reasoning where structured extraction alone is insufficient. The pipeline produces a canonical JSON record per document, with confidence scores at the field level.

Outcome

- Structured data: typed fields, line-item tables, key-value pairs, codes
- Semi-structured data: notes, narratives, condition observations
- Visual entities: stamps, signatures, photo evidence, redaction marks
- Confidence scores for every field, driving routing decisions downstream

4.3 Data Lake Creation

Microsoft Fabric and OneLake materialize the unified data lake. Every captured document and every extraction outcome is persisted with full lineage, classification, and access governance.

Outcome

- Bronze layer: raw documents and capture metadata (immutable, audit-grade)
- Silver layer: validated extractions, confidence scores, links to source documents
- Gold layer: business-ready entities — shipment, consignment, billing event, exception

- Direct Lake access for Power BI dashboards and analytical workloads — no separate warehouse copy
- Native Microsoft Purview cataloging for sensitive-data classification and lineage

4.4 Data Insights During Processing

Insight generation happens in-flight, not as a batch report at month-end. As documents land, anomaly detection, business-rule evaluation, and cross-document reconciliation run continuously.

Outcome

- Volume and SLA dashboards refreshed in near real-time
- Exception forecasting based on inbound document mix
- Compliance signals (sanctions, dangerous-goods, duty exposure) raised at point of capture
- Customer-facing insights: shipment status, billing reconciliation, anomaly alerts
- Operator-facing insights: queue depth by exception class, SME availability, expected handling time

5. Workflow Routing and Next Best Action

Routing is the connective tissue of the engine. Logic Apps and Power Automate orchestrate the flow of work between automated processing, Conduent operators, customer SMEs, and downstream systems — guided by confidence scores, business rules, and SLA timers.

5.1 Routing dimensions

- **Confidence-based.** High-confidence extractions (typically 90%+ across all fields) are auto-posted to the customer system of record — straight-through processing. Medium-confidence extractions are routed to a Conduent assisted-review queue. Low-confidence extractions or classifier ambiguity are routed to SME adjudication.
- **Rule-based.** Customs anomalies, dangerous-goods declarations, sanctions screen hits, and duty-rate variances are routed to specialist queues regardless of confidence. The reasoning trace from Azure OpenAI accompanies the routing decision.
- **Cross-document validation.** When linked documents (invoice, AWB, BOL, packing list) are reconciled and inconsistencies are found, the bundle is held and escalated as a unit rather than processed piecemeal.
- **Customer-specific.** Each customer has a configurable routing policy — different SLA tiers, different SME assignments, different escalation paths.

5.2 Next-best-action generation

Every routed work item carries an Azure OpenAI-generated next-best-action narrative — a plain-language explanation of why the item is in the queue, what is wrong, and what the recommended resolution is. This is not advisory only; the operator can accept the recommendation with a single click, and the resolution is logged for active learning.

Examples of next-best-action narratives include:

- 'AWB 020-12345678 has a declared weight of 4,500 kg but the corresponding packing list sums to 3,200 kg. Recommended action: hold for shipper confirmation; auto-draft email below.'
- 'Customs declaration shows HS code 8517.12 (mobile phones) but the commercial invoice describes textile goods. Recommended action: route to customs broker for re-classification.'
- 'PoD signature is illegible and condition note reads 'one carton damaged'. Recommended action: open exception case; auto-link photo evidence; notify consignee.'

5.3 Alerts and escalation

Logic Apps maintains SLA timers per document, per customer, per queue. As thresholds approach, alerts are issued via Microsoft Teams to the Conduent operations lead and, where contractually agreed, to the customer SME. Alerts include the next-best-action narrative so that recipients can act immediately rather than triage further.

6. Conduent and Customer SME Engagement Model

The success of the engine depends on a clean, well-instrumented engagement model between Conduent operators and customer-side subject matter experts. The Microsoft platform provides the connective tissue.

6.1 Conduent operations

Conduent operators handle:

- Tier-1 exception queues — assisted review of medium-confidence extractions
- Document re-capture workflows when image quality is insufficient
- Duplicate detection and resolution
- First-line customer communications and inquiry handling
- SLA monitoring and operational reporting
- Pattern reporting back to the customer for upstream improvement (e.g., 'these three shippers consistently submit photographs that fail capture quality')

6.2 Customer subject matter experts

Customer SMEs handle:

- Tier-2 and tier-3 adjudication — customs classification, dangerous-goods reasoning, duty disputes
- Business-rule definition and evolution — what 'normal' looks like for their operation
- Onboarding new shippers, lanes, and document types — providing initial example sets
- Final approval on regulated submissions
- Periodic review of analytics and trends surfaced from the data lake

6.3 Shared queue and Copilot agents

Conduent operators and customer SMEs share a unified work queue surfaced in Microsoft Teams via Copilot Studio agents. Each side sees the items relevant to their tier, with full context, source documents, validation history, and the AI-generated next-best-action narrative.

Routing between tiers is automatic and bidirectional — a Conduent operator can escalate to the customer SME with a single action; a customer SME can return an item with annotation back to the Conduent queue. Every transition is logged, and every correction feeds the active-learning loop.

6.4 Active learning and continuous improvement

SME corrections are captured as labeled training data. On a weekly cadence, the Document Intelligence custom models are retrained against the accumulated correction set; the Azure OpenAI prompts are refined based on emerging exception patterns; and the routing thresholds are recalibrated based on observed accuracy by document type and customer.

Continuous improvement is operationalized — not a periodic project. Quality, coverage, and economics of the pipeline improve every week.

The operating model in one sentence

Conduent operations runs the volume; the customer SMEs anchor the judgment; the Microsoft AI platform amplifies both — and every interaction makes the next one faster.

7. Conversion Plan — From Legacy OCR to Azure AI

The shift from the legacy OCR estate to the Azure AI engine is delivered in a phased, parallel-run pattern that protects production volume while compressing the timeline. The plan below assumes a single anchor customer with the document mix typical of a large freight or airline cargo operation.

7.1 Phased plan

Phase 1 — Discover and Assess (2 to 3 weeks)

- Inventory active document types, vendor layouts, and processing volumes
- Capture baseline OCR accuracy, exception rates, and onboarding cycle times
- Catalog upstream and downstream integration points (ERP, TMS, customs systems)
- Identify two or three high-volume document types to anchor the pilot
- Define accuracy and economic targets for the pilot

Phase 2 — Foundation (3 to 4 weeks)

- Stand up Azure landing zone with hub-and-spoke networking, identity, key management
- Provision Microsoft Fabric workspace and OneLake medallion architecture
- Establish Microsoft Purview catalog and sensitivity-label scheme
- Stand up Document Intelligence and Azure OpenAI in the target region with private endpoints
- Establish CI/CD pipelines (GitHub Actions or Azure DevOps) with infrastructure-as-code

Phase 3 — Pilot (4 to 6 weeks)

- Build extractors for the anchor document types — typically AWBs, commercial invoices, and weigh bills
- Train custom Document Intelligence models on representative samples (1,000–3,000 per type)
- Implement workflow routing in Logic Apps with confidence-based and rule-based dispatch
- Deploy operator and SME Copilot agents in Teams
- Run a controlled pilot on 10,000–25,000 documents per anchor type, comparing accuracy and economics against the OCR baseline
- Target: 90%+ STP on anchor types; 50%+ reduction in cost per document

Phase 4 — Parallel Run (4 weeks)

- Both pipelines processing live volume on the anchor document types
- Daily reconciliation of extraction outputs and exception classification
- Customer SLA continuity verified end-to-end
- Confidence threshold tuning based on observed error rates
- Decision gate: green-light scale-out

Phase 5 — Scale-Out (8 to 12 weeks)

- Onboard remaining document types in waves of three to four per two-week sprint
- Each wave: sample collection, custom model training, prompt engineering, regression testing, parallel run, cutover
- Power BI dashboards and customer-facing analytics released in parallel
- Active-learning loop in continuous operation

Phase 6 — Cutover and Decommission (2 to 3 weeks)

- Final OCR templates retired; legacy pipeline placed in read-only archival mode
- Historical data migrated into OneLake bronze layer for analytics continuity
- Operations playbooks, runbooks, and SME guides finalized
- Lessons-learned capture and customer business case formalization

End-to-end timeline

Discover (3 weeks) + Foundation (4 weeks) + Pilot (6 weeks) + Parallel Run (4 weeks) + Scale-Out (12 weeks) + Cutover (3 weeks) — totaling 32 weeks, or roughly 7.5 months. Aggressive engagements with strong executive sponsorship and a favorable document mix have completed in 4 to 6 months. This compares with 12 to 18 months for a traditional rebuild on legacy stacks.

8. Accelerated AI-Driven SDLC

The aggressive timeline above is achievable because the entire delivery is itself executed on AI-augmented engineering tools. This section enumerates the productivity multipliers and the practical engineering cadence.

8.1 AI-augmented engineering practices

- **GitHub Copilot for application code.** Extractor logic, validation rules, integration adapters, Logic Apps custom connectors, and Power BI calculations are authored with Copilot pair-programming. Microsoft published benchmarks indicate 40-60% velocity uplift on greenfield code.
- **GitHub Copilot Workspace and Copilot in Azure for infrastructure.** Bicep / Terraform for Azure landing zone, Fabric workspace, and AI services authored with infrastructure copilots. Configuration drift detection and remediation are AI-assisted.
- **AI-generated unit and integration tests.** Test cases for extractors and validation rules are generated against real-document samples, dramatically expanding coverage compared with hand-authored test suites.
- **Synthetic test data generation.** For document types where real samples are scarce or sensitive, Azure OpenAI is used to generate variant samples for regression testing — within governance controls.
- **AI-assisted prompt engineering.** Prompt patterns for Azure OpenAI extractors are iterated against scored test sets — Copilot suggests prompt revisions and the engineering team selects, with versioning in source control.
- **Auto-generated user stories and acceptance criteria.** Document samples are analyzed by Copilot to draft initial user stories, acceptance criteria, and edge-case lists for the engineering team to refine.
- **AI-generated documentation.** Runbooks, operator guides, and customer-facing release notes are drafted from change history and configuration metadata.

8.2 Engineering cadence

With the AI-augmented practices in place, the practical engineering cadence is:

1. Two-week sprints aligned to the phased plan above
2. Each sprint produces extractors, routing rules, dashboards, and operator agents for one or more new document types
3. Daily AI-assisted code review through GitHub Copilot suggestions and security scanning
4. Continuous deployment to non-production with promotion gates to parallel-run and production
5. Weekly retraining cadence for custom Document Intelligence models, with automated regression validation

8.3 Why this is fast

Three structural factors compound to enable the accelerated timeline:

- Foundation models replace template engineering — the most labor-intensive task in legacy OCR conversions disappears
- Microsoft platform integration is pre-wired — Document Intelligence, Azure OpenAI, Fabric, Logic Apps, and Teams compose natively without custom integration
- AI-augmented engineering compresses every craft step — code, tests, infrastructure, documentation, and prompts are all accelerated by Copilot tooling

Modernization not as a rebuild — as an upgrade

The combination of Azure AI services, the Microsoft platform stack, and AI-augmented engineering reframes this from a multi-year systems modernization to a structured, six-month operational uplift. The result is an AI-native document operation that grows sharper with use, scales to absorb new customers in days, and turns the document stream into a strategic data asset.

9. Closing — From Solution Map to Execution

This solution map provides the architectural blueprint for the next-generation Conduent document processing engine. The companion Volume 1 establishes the strategic case; Volume 3 distills the architecture into an executive briefing deck.

Recommended next steps:

1. Joint discovery workshop — Conduent and customer leadership — to validate document mix, anchor types, and target metrics
2. Establish Azure landing zone and Fabric workspace in parallel
3. Stand up the four-to-six-week pilot on the agreed anchor document types
4. Convene the executive review at end-of-pilot to confirm scale-out commitment
5. Execute the phased scale-out and cutover on the timeline articulated in Section 7

The path is clear. The technology is available. The operating model is proven. What remains is the disciplined execution of a well-scoped pilot — and the conversion of cargo document processing from a high-cost, template-bound operation into the AI-native intelligence engine the industry now requires.